

Beyond Reconstruction: Evaluating Representational Fidelity in Continuous Neural Representations

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Abstract

Continuous neural representations (CNRs) are increasingly used for medical image analysis, yet their evaluation remains dominated by reconstruction metrics such as the Dice coefficient. While these metrics quantify output accuracy, they do not directly assess the quality of the learned representation. This study investigates whether models with statistically equivalent reconstruction performance can preserve anatomically meaningful information with substantially different levels of representational fidelity. A quantitative evaluation framework is proposed that combines geometric, latent-space, and representational similarity analyses into a unified Representational Fidelity Index (RFI). The framework is validated on two independent medical imaging datasets: retinal vessels (DRIVE) and gland ground truth masks (GLAS), using multiple SIREN variants trained under identical optimization and evaluation protocols. Reconstruction equivalence is formally established (Two One-Sided Tests for GLAS, Bonferroni-corrected non-inferiority for DRIVE), ensuring that comparisons of representational fidelity are not confounded by differences in reconstruction accuracy. Despite statistically indistinguishable reconstruction performance, the models exhibit consistent differences in geometric fidelity, latent representation structure, representational similarity, and composite RFI. These differences are deterministic with respect to random initialization and robust to variations in the composite index formulation. The results demonstrate that reconstruction fidelity provides only a partial characterization of continuous neural representations and that direct evaluation of representational fidelity offers complementary information beyond reconstruction accuracy. Representational fidelity should therefore be considered alongside reconstruction fidelity when evaluating continuous neural representations for medical image analysis.

Reproducibility. Complete experimental implementations are publicly available for the [DRIVE](#) and [GLAS](#) datasets.

1. Introduction

Continuous neural representations (CNRs) model medical images and anatomical structures as continuous functions that can be queried at arbitrary spatial locations. Representative formulations include implicit neural representations (SIRENs) [1], signed distance functions (SDFs) [2], occupancy networks [3], and neural radiance fields (NeRFs) [4]. These representations provide a compact, resolution-independent description of anatomy and have demonstrated strong performance across a variety of medical imaging tasks, including image reconstruction [7], segmentation [8], registration [9], and geometric modelling [10]. Their ability to capture fine anatomical detail with sub-voxel precision makes them an attractive alternative to discrete image representations.

Despite these advances, evaluation of continuous neural representations remains centred on reconstruction accuracy. Model quality is typically assessed using metrics such as the Dice similarity coefficient (Dice), Hausdorff distance (HD95), and average surface distance (ASD), which quantify the agreement between reconstructed outputs and reference annotations [11, 13, 14]. Although indispensable for measuring reconstruction quality, these metrics characterise only the fidelity of the decoded output and provide limited insight into the information preserved within the learned representation itself. Consequently, reconstruction accuracy is often treated as a surrogate for representation quality, even though reconstruction fidelity and representational fidelity capture fundamentally different aspects of a learned model [15].

This distinction is scientifically important because reconstruction fidelity and representational fidelity are not necessarily equivalent. Reconstruction fidelity measures how accurately a decoded representation reproduces the target anatomy, whereas representational fidelity reflects how faithfully anatomically meaningful information is encoded within the underlying continuous representation. Two models may therefore achieve statistically indistinguishable reconstruction performance while learning substantially different internal representations. Such differences can influence robustness to perturbations [30], interpolation behaviour, transferability across tasks, and other downstream analyses that rely directly on the learned representation rather than solely on the reconstructed output [16]. If only reconstruction accuracy is used for evaluation, these representational differences may remain entirely undetected.

Representation analysis has attracted considerable attention in deep learning through studies of representation similarity, centred kernel alignment (CKA) [21], singular vector canonical correlation analysis (SVCCA) [20], intrinsic dimensionality, effective rank, and representation collapse [22, 23]. These techniques have primarily been developed to understand representations learned by conventional deep neural networks. Research on continuous neural representations, by contrast, has focused predominantly on reconstruction performance, with comparatively little attention devoted to analysing the properties of the learned representations themselves [1, 2, 3]. Consequently, the application of representation analysis to continuous neural representations in medical imaging remains largely unexplored. A systematic evaluation of whether models with statistically equivalent reconstruction performance also preserve anatomically meaningful information with equivalent representational fidelity has not yet been reported.

Motivated by this gap, the present study investigates the following hypothesis:

Continuous neural representations that achieve statistically equivalent reconstruction performance can nevertheless differ substantially in their ability to preserve anatomically meaningful information.

To test this hypothesis, a quantitative framework is developed for evaluating representational fidelity independently of reconstruction fidelity. The framework integrates geometric fidelity, latent-space richness, and representational similarity analyses into a composite Representational Fidelity Index (RFI). The framework is validated on two independent medical imaging datasets: retinal vessel images (DRIVE) and gland ground truth masks (GLAS). In both experiments, multiple SIREN variants are trained under controlled conditions, and reconstruction equivalence is formally established before representational comparisons are performed. Statistical rigour is ensured through bootstrap confidence intervals, equivalence testing, multiple-comparison correction, robustness analyses, and repeated runs with different random seeds.

The main contributions of this work are as follows:

1. A quantitative framework for evaluating representational fidelity independently of reconstruction fidelity in continuous neural representations is presented.
2. A composite Representational Fidelity Index (RFI) is introduced, integrating complementary geometric, latent, and similarity characteristics into a unified quantitative measure.

3. The framework is validated across two independent medical imaging datasets, demonstrating that continuous neural representations with statistically equivalent reconstruction performance can exhibit substantially different representational fidelity.
4. Comprehensive statistical analysis is performed, including equivalence testing, bootstrap confidence intervals, Holm-corrected multiple comparisons, sensitivity analyses, and repeated random-seed experiments.
5. The results provide empirical evidence that representational fidelity constitutes an independent and complementary dimension of model quality beyond conventional reconstruction metrics, motivating its explicit evaluation in future studies.

The remainder of this paper is organised as follows. Section 2 reviews related work. Section 3 presents the proposed methodology. Section 4 describes the experimental design and evaluation protocol. Section 5 reports the experimental results from both datasets. Section 6 discusses the implications and limitations of the framework. Finally, Section 7 concludes the paper.

2. Related Work

2.1 Continuous Neural Representations

Continuous neural representations (CNRs) model spatial signals as continuous functions parameterised by neural networks, enabling querying at arbitrary coordinates. SIREN [1] demonstrated that periodic activation functions allow stable learning of high-frequency detail, while DeepSDF [2] and Occupancy Networks [3] established implicit representations based on signed distance functions and occupancy fields for continuous surface modelling. NeRF [4] further showcased the expressive power of implicit neural representations for complex 3D scenes. Building on these advances, CNRs have been successfully applied to a broad range of medical imaging tasks, including reconstruction, segmentation, registration, and anatomical shape analysis [7, 8, 9, 10]. While these studies confirm the effectiveness of CNRs for anatomical modelling, they have focused primarily on reconstruction performance rather than on the fidelity of the learned representations themselves.

2.2 Evaluation of Medical Image Segmentation

Medical image segmentation is conventionally evaluated using reconstruction-oriented metrics such as the Dice coefficient [11], Hausdorff distance [12], and average surface distance [14]. These metrics quantify spatial overlap and boundary agreement between predicted and reference segmentation and thus serve as the standard criteria for assessing reconstruction performance [13]. Although indispensable for measuring reconstruction accuracy, they characterise only the quality of the decoded output and provide limited insight into the information encoded within the learned representation. This limitation becomes increasingly important for continuous neural representations, where the learned representation itself often forms the basis for downstream applications such as anatomical modelling, interpolation, registration, and quantitative shape analysis [18]. Consequently, high reconstruction accuracy alone does not guarantee that a learned representation faithfully preserves anatomically meaningful information.

2.3 Analysis of Learned Representations

Understanding learned representations has become an important research direction in deep learning. Numerous analytical techniques have been developed to characterise representation similarity, information content, and structural complexity beyond predictive performance, including centred kernel alignment (CKA) [21],

singular vector canonical correlation analysis (SVCCA) [20], intrinsic dimensionality analysis, effective rank, and studies of representation collapse [22, 23]. Complementary work on implicit neural representations has also identified spectral bias, where networks preferentially learn low-frequency signals before high-frequency details [24, 25, 1]. Collectively, these studies show that models with similar predictive performance may learn substantially different internal representations. However, these analytical approaches have been developed primarily for conventional deep neural networks and have rarely been applied to evaluate the quality of continuous neural representations in medical imaging.

2.4 Research Gap

Although continuous neural representations and representation analysis have both advanced rapidly, they have largely evolved independently. Existing studies of CNRs continue to evaluate models predominantly through reconstruction-based metrics, whereas research on learned representations has focused on conventional deep networks. Consequently, the relationship between reconstruction fidelity and representational fidelity in continuous neural representations remains poorly understood. In particular, it has not been systematically investigated whether models that achieve statistically equivalent reconstruction performance also preserve anatomically meaningful information with equivalent representational fidelity. The present study addresses this gap by introducing a quantitative framework for evaluating representational fidelity independently of reconstruction fidelity and validating the framework through controlled experiments on two independent medical imaging datasets.

3. Methods

3.1 Overall Research Framework

The overall experimental framework is designed to evaluate representational fidelity independently of reconstruction fidelity in continuous neural representations (CNRs). As illustrated in Figure 1, the workflow begins by learning continuous neural representations from medical image data using SIREN-based implicit neural networks. Conventional reconstruction metrics are first employed to establish reconstruction performance and, where appropriate, statistical equivalence between competing models. Once reconstruction differences have been controlled, the learned representations are analyzed directly through complementary geometric, latent-space, and representational similarity analyses. These measurements are subsequently integrated into a composite Representational Fidelity Index (RFI), enabling quantitative comparison of models beyond reconstruction accuracy alone. Finally, reconstruction fidelity and representational fidelity are jointly analyzed to determine whether statistically equivalent reconstruction performance necessarily implies equivalent preservation of anatomically meaningful information, thereby providing a direct evaluation of the central research hypothesis.

3.2 Problem Formulation

Let $\mathcal{D} = \{(M_i)\}_{i=1}^N$ be a set of binary ground truth masks $M_i : \Omega \rightarrow \{0, 1\}$ over a spatial domain $\Omega \subset \mathbb{R}^2$. A continuous neural representation $f_\theta : \Omega \rightarrow \mathbb{R}$ is trained to regress the signed distance function (SDF) derived from M_i [17]:

$$SDF(M_i)(x) = \begin{cases} +d(x, \partial M_i) & \text{if } x \in M_i \\ -d(x, \partial M_i) & \text{otherwise} \end{cases} \quad (1)$$

where $d(x, \partial M_i)$ is the Euclidean distance to the nearest boundary point. The SDF is normalised to have a maximum absolute value of 1. The trained network f_θ implicitly encodes the anatomy; the zero level set

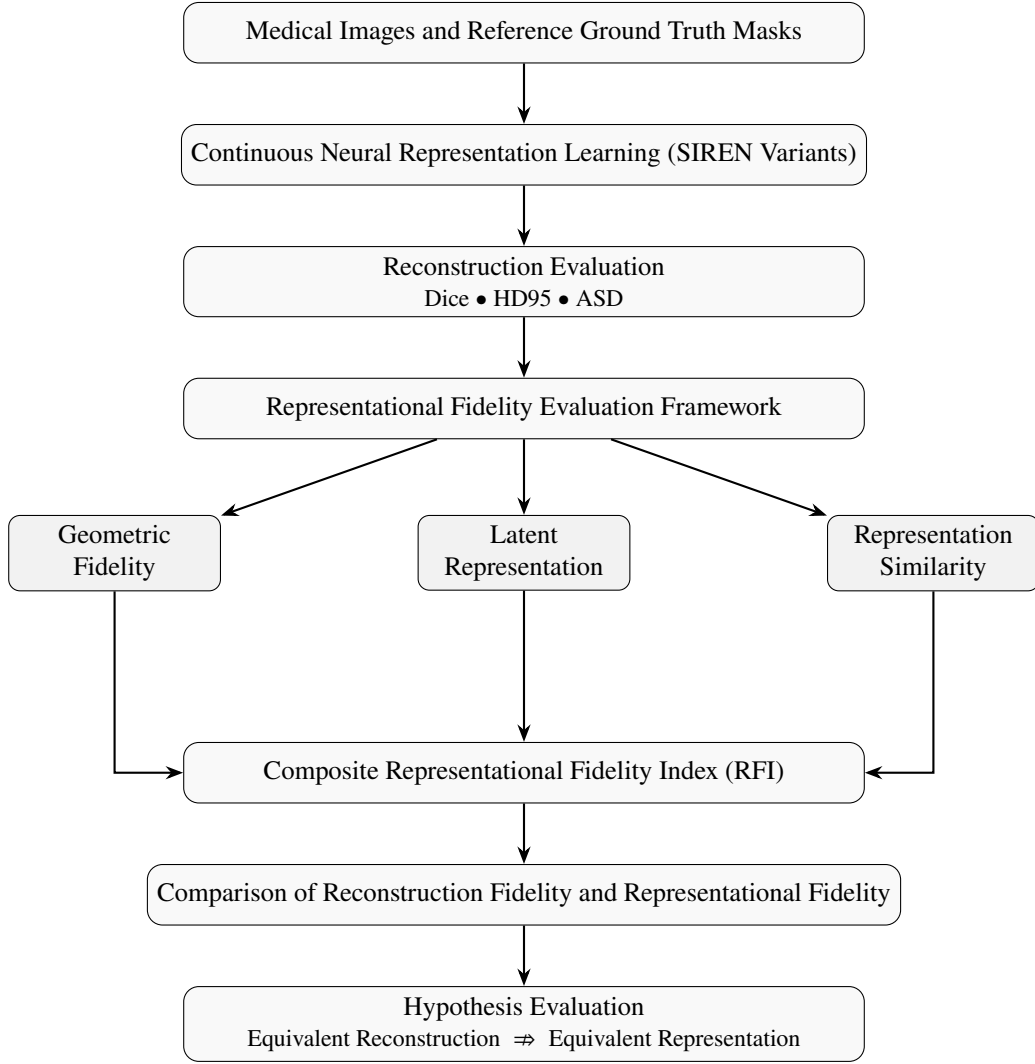


Figure 1: Overview of the proposed research framework. Continuous neural representations are first learned from medical image data and evaluated using conventional reconstruction metrics. After controlling for reconstruction performance, the learned representations are analyzed through complementary geometric, latent-space, and representation similarity measures, which are integrated into a composite Representational Fidelity Index (RFI). The resulting framework enables direct comparison between reconstruction fidelity and representational fidelity to evaluate whether statistically equivalent reconstruction performance corresponds to equivalent preservation of anatomically meaningful information.

$\{x : f_\theta(x) = 0\}$ defines the reconstructed boundary.

Reconstruction fidelity quantifies the agreement between the reconstructed mask $\hat{M}_i = \mathbb{K}(f_\theta(x) > 0)$ and the ground truth M_i , measured by Dice [11], HD95 [12], and ASD [14].

Representational fidelity quantifies the extent to which f_θ preserves anatomically meaningful information beyond the mere boundary location. This is captured through geometric consistency (gradient and curvature errors near the zero level set), latent-space richness (effective rank, spectral entropy, participation ratio, stable rank), and representational similarity (CKA). A composite Representational Fidelity Index (RFI) combines these into a single scalar (lower RFI indicates better fidelity).

3.3 Continuous Representation Models

All experiments employ the SIREN architecture [1], which uses sinusoidal activation functions $\sin(\omega_0 \cdot)$ with a user-defined frequency parameter ω_0 . The base network consists of 4 hidden layers with 128 units each and a final linear layer. The input is a 2D coordinate $x \in [-1, 1]^2$.

For the DRIVE experiments, SIREN variants with different hidden dimensions are compared: 112, 128, 160, and 192 units, all with $\omega_0 = 30$ and no regularisation. This design allows identification of a pair of models that are matched in reconstruction performance but differ in capacity.

For the GLAS experiments, the hidden dimension is fixed at 128 and the frequency ω_0 and training regularisation are systematically varied, resulting in four variants:

- **SIREN-128** ($\omega_0 = 30$, no regularisation)
- **SIREN-128-Eikonal** ($\omega_0 = 30$, eikonal regularisation $\lambda = 10^{-4}$)
- **SIREN-128-LowOmega** ($\omega_0 = 15$, no regularisation)
- **SIREN-128-Jacobian** ($\omega_0 = 30$, Jacobian norm regularisation $\lambda = 10^{-4}$)

All models are trained with the Adam optimiser [26] using a learning rate of 5×10^{-5} . Training proceeds for 5000 epochs (DRIVE) or 2000 epochs (GLAS) with early stopping (patience of 500 or 800 evaluations, respectively). The coordinate grid spans $[-1, 1]^2$ and is fully loaded onto the GPU for efficient batch training.

3.4 Representational Fidelity Evaluation

Representational fidelity is evaluated from three complementary perspectives.

3.4.1 Geometric Fidelity

Geometric fidelity measures how accurately the learned SDF field reproduces the local and global geometry of the true anatomy near the boundary [17].

- **Gradient error** (GradErr): angular deviation (in degrees) between the predicted SDF gradient and the true gradient, averaged over a narrow band around the zero level set.
- **Curvature error** (CurvErr): relative mean absolute error of the mean curvature computed from the SDF, evaluated on the boundary.
- **Euler characteristic error** (EulerCharErr): weighted absolute difference between the Euler characteristic of the ground-truth mask and that obtained by thresholding the predicted SDF at 50 level sets [19]. The Euler characteristic provides a topological invariant that captures the number of connected components and holes.
- **Vessel width error** (WidthErr, DRIVE only): mean absolute difference in vessel width estimated from 1D cross-sections of the SDF along the normal direction at skeleton points [5].

3.4.2 Latent Representation Richness

To quantify the information content of the hidden activations, for each layer the covariance matrix $\Sigma = \frac{1}{N-1} \mathbf{H}^\top \mathbf{H}$ of the centred feature matrix $\mathbf{H} \in \mathbb{R}^{N \times d}$ is computed, where N is the number of spatial query

points and d is the feature dimension. Let $\lambda = (\lambda_1, \dots, \lambda_d)$ be the eigenvalues of Σ in descending order [22]. From these eigenvalues, the following metrics are derived:

$$\text{Spectral entropy} \quad H(\lambda) = - \sum_i \bar{\lambda}_i \log \bar{\lambda}_i, \quad \bar{\lambda}_i = \lambda_i / \sum_j \lambda_j \quad (2)$$

$$\text{Effective rank} \quad \text{ER} = \exp(H(\lambda)) \quad (3)$$

$$\text{Stable rank} \quad \text{SR} = \frac{\sum_i \lambda_i}{\lambda_{\max}} \quad (4)$$

$$\text{Participation ratio} \quad \text{PR} = \frac{(\sum_i \lambda_i)^2}{\sum_i \lambda_i^2} \quad (5)$$

These metrics capture the effective dimensionality of the representation; higher values indicate richer, more diverse feature utilisation. The **coordinate sensitivity**, defined as the mean Frobenius norm of the Jacobian $\partial f_\theta / \partial x$, is also computed as a measure of spatial sharpness.

3.4.3 Representational Similarity

Centred kernel alignment (CKA) [21] is used to compare the hidden representations of different models or layers. Given two centred feature matrices $\mathbf{H}_1, \mathbf{H}_2$, the linear CKA is

$$\text{CKA}(\mathbf{H}_1, \mathbf{H}_2) = \frac{\|\mathbf{H}_1^\top \mathbf{H}_2\|_F^2}{\|\mathbf{H}_1^\top \mathbf{H}_1\|_F \|\mathbf{H}_2^\top \mathbf{H}_2\|_F}. \quad (6)$$

Pairwise CKA is computed between models for each layer, and intra-model CKA is computed between consecutive layers to assess representation compression.

3.4.4 Composite Representational Fidelity Index (RFI)

To obtain a single summary statistic, each geometric error metric is first z -scored using the SIREN model family as reference (mean and standard deviation computed over all SIREN runs). The **GeoIndex** is the mean of the z -scored geometric errors. Similarly, each latent richness metric is z -scored across all models, and the **LatentIndex** is the negative mean of the z -scored richness metrics (so that higher richness gives a more negative LatentIndex). The composite index is

$$\text{RFI} = \frac{\text{GeoIndex} + \text{LatentIndex}}{2}. \quad (7)$$

Lower (more negative) RFI indicates better representational fidelity. Sensitivity to the weighting of the two components is examined in Section 5.2.3.

3.5 Statistical Analysis

A rigorous statistical pipeline is employed:

- Bootstrap 95% confidence intervals (500–2000 resamples) for all means and paired differences [27].
- Wilcoxon signed-rank tests for paired model comparisons, corrected with Holm’s method [28].
- Equivalence testing using TOST for Dice with a predefined bound of ± 0.02 (GLAS) or Bonferroni-corrected non-inferiority (DRIVE) [29].
- Spearman rank correlations between Dice and each fidelity metric, also Holm-corrected.

- Sensitivity analysis of RFI to varying the GeoIndex/LatentIndex weighting (0.3/0.7, 0.5/0.5, 0.7/0.3).
- Deterministic consistency assessment: when paired differences are identical across all samples, formal hypothesis tests become inapplicable; instead, bootstrap CIs for differences are reported.

All experiments are run with multiple random seeds (3 for DRIVE, 2 for GLAS), and all metrics are averaged per image before further analysis.

4. Experimental Setup

4.1 Datasets

Two publicly available medical imaging datasets are used:

- **DRIVE (Digital Retinal Images for Vessel Extraction)** [5]: 20 retinal fundus images with expert vessel annotations, resized to 128×128 pixels.
- **GLAS (Gland Segmentation in Colon Histology Images)** [6]: 20 non-degenerate H&E-stained images of colorectal glands, resized to 256×256 pixels.

Both datasets provide binary ground truth masks with complex, fine-scale structures. Normalised SDFs are computed from each mask and serve as the regression target. In the DRIVE experiment, occupancy and ReLU-SDF baselines were also trained but failed to produce meaningful reconstructions and are excluded from the main analysis.

4.2 Training and Evaluation

All models are trained to regress the full-image SDF. The Adam optimiser [26] ($\text{lr} = 5 \times 10^{-5}$) is used with early stopping. For DRIVE, training runs for up to 5000 epochs with patience 500; for GLAS, up to 2000 epochs with patience 800. After training, each model predicts the SDF over the full grid. The binary mask is obtained by thresholding at zero. Reconstruction metrics (Dice, HD95, ASD) are computed against the ground truth. Geometric fidelity metrics are calculated directly from the predicted SDF and ground-truth SDF. Latent metrics are derived from the hidden activations of the final layer (or all layers for CKA). CKA is computed using all spatial query points. The composite RFI is calculated per image-model pair, and final statistics are aggregated across images.

5. Results

5.1 Experiment 1: DRIVE Retinal Vessels

5.1.1 Reconstruction Performance and Equivalence

Table 1 summarises the reconstruction metrics for the four SIREN variants. SIREN-128 and SIREN-112 achieved nearly identical Dice scores (0.722 vs. 0.719) with overlapping confidence intervals. Wilcoxon signed-rank tests with Bonferroni correction confirmed that no reconstruction metric differed significantly between this pair ($p > 0.05$ for Dice, HD95, ASD). All other pairs had at least one significantly different reconstruction metric, leaving SIREN-128 and SIREN-112 as the only models with statistically equivalent reconstruction performance.

Table 1: Reconstruction metrics for DRIVE SIREN variants (mean [95% CI]).

Model	Dice	HD95 (pixels)	ASD (pixels)
SIREN-112	0.719 [0.706, 0.733]	5.83 [5.36, 6.29]	0.616 [0.561, 0.664]
SIREN-128	0.722 [0.708, 0.736]	5.81 [5.36, 6.24]	0.610 [0.558, 0.658]
SIREN-160	0.775 [0.764, 0.785]	4.74 [4.32, 5.16]	0.466 [0.422, 0.511]
SIREN-192	0.812 [0.803, 0.821]	3.94 [3.62, 4.28]	0.362 [0.327, 0.396]

5.1.2 Representational Fidelity of the Matched Pair

Despite equivalent reconstruction, SIREN-128 and SIREN-112 showed clear differences in representational fidelity (Table 2). The composite RFI was -0.234 for SIREN-128 versus -0.145 for SIREN-112 (Cohen’s $d_z = -2.65$, $p < 0.0001$). The LatentIndex diverged even more strongly (-0.286 vs. -0.150 , $d_z = -9.16$). Geometric metrics also favoured the larger model: gradient error 30.44° vs. 30.55° ($p = 0.040$), curvature error 0.955 vs. 0.960 ($p = 0.033$), and vessel width error 2.60 vs. 2.67 pixels ($p = 0.097$, trend level). Euler characteristic error did not differ significantly.

Table 2: Representational fidelity for the matched DRIVE pair (mean [95% CI]).

Metric	SIREN-128	SIREN-112
RFI	-0.234 [-0.301 , -0.170]	-0.145 [-0.210 , -0.080]
LatentIndex	-0.286 [-0.295 , -0.276]	-0.150 [-0.161 , -0.139]
GeoIndex	-0.182 [-0.326 , -0.051]	-0.141 [-0.280 , -0.005]
GradErr ($^\circ$)	30.44 [29.74 , 31.18]	30.55 [29.84 , 31.34]
CurvErr	0.955 [0.946 , 0.966]	0.960 [0.950 , 0.970]
WidthErr (pixels)	2.60 [2.32 , 2.89]	2.67 [2.42 , 2.96]
EulerCharErr	170.1 [157.1 , 182.4]	170.2 [156.6 , 182.1]
EffectiveRank	84.84 [84.32 , 85.37]	74.02 [73.54 , 74.48]
SpectralEntropy	4.441 [4.435 , 4.447]	4.304 [4.297 , 4.310]
ParticipationRatio	61.30 [60.74 , 61.81]	53.58 [53.05 , 54.09]

5.1.3 Representational Similarity

Layer-wise CKA between SIREN-128 and SIREN-112 declined from 0.719 (layer 1) to 0.380 (layer 2) and 0.236 (layer 3), indicating that the two models develop increasingly distinct internal representations despite converging to almost identical output masks. Intra-model CKA also differed: SIREN-128 exhibited layer-to-layer CKA values of 0.639 and 0.647 , whereas SIREN-112 yielded 0.667 and 0.641 , suggesting a slightly more compressed representation in the smaller model.

5.1.4 Robustness to Input Perturbations

When Gaussian noise was added to the input coordinates [30], SIREN-128 showed a 70% drop in Dice (from 0.722 to 0.215 at noise std 0.05), while SIREN-112 dropped to 0.069 (a 90% relative reduction from its baseline of 0.719). This indicates that the richer latent representation of the larger model may confer greater stability under input perturbations, although further study is needed to establish this link definitively.

5.2 Experiment 2: GLAS Gland Segmentation

5.2.1 Reconstruction Performance and Equivalence

All four SIREN-128 variants achieved excellent Dice scores (0.986–0.995), as shown in Table 3. TOST confirmed that **all six model pairs were statistically equivalent in Dice** within ± 0.02 (Holm-adjusted $p < 0.0001$ for every pair). HD95 and ASD were also low, though LowOmega exhibited moderately higher boundary errors.

Table 3: Reconstruction metrics for GLAS (mean [95% CI]).

Model	Dice	HD95 (pixels)	ASD (pixels)
SIREN-128	0.995 [0.994, 0.996]	1.85 [1.00, 3.56]	0.151 [0.086, 0.264]
SIREN-128-Eikonal	0.995 [0.994, 0.996]	2.15 [1.00, 4.41]	0.184 [0.094, 0.348]
SIREN-128-LowOmega	0.986 [0.983, 0.989]	5.43 [1.54, 10.90]	0.591 [0.270, 1.002]
SIREN-128-Jacobian	0.995 [0.994, 0.996]	2.08 [1.00, 4.25]	0.177 [0.087, 0.335]

5.2.2 Representational Fidelity

Geometric fidelity metrics (Table 4) revealed that LowOmega produced systematically larger gradient errors (12.45° vs. $\sim 10.5^\circ$) and moderately worse curvature. The most striking divergence appeared in the latent metrics (Table 5): LowOmega’s effective rank dropped to 43.5, a 41% reduction compared to the $\omega_0 = 30$ models. Participation ratio fell from ~ 51 to 28, and spectral entropy from 4.31 to 3.77. Coordinate sensitivity remained comparable, indicating that the impoverished representation is not simply a result of a smoother field.

Table 4: Geometric fidelity for GLAS (mean [95% CI]).

Model	GradErr ($^\circ$)	CurvErr	EulerCharErr
SIREN-128	10.58 [10.07, 11.14]	1.010 [0.991, 1.031]	5.94 [3.71, 8.38]
SIREN-128-Eikonal	10.58 [10.00, 11.23]	1.007 [0.988, 1.028]	6.07 [3.76, 8.33]
SIREN-128-LowOmega	12.45 [11.52, 13.27]	1.042 [1.022, 1.058]	6.19 [3.93, 8.93]
SIREN-128-Jacobian	10.51 [9.96, 11.03]	1.004 [0.987, 1.024]	5.95 [3.66, 8.27]

Table 5: Latent representation richness for GLAS (mean [95% CI]).

Model	Eff.Rank	SpectralEnt.	StableRank	Part.Ratio	CoordSens
SIREN-128	74.44 [73.98, 75.14]	4.310 [4.303, 4.318]	20.30 [20.06, 20.58]	51.73 [51.24, 52.36]	1.886 [1.529, 2.282]
SIREN-128-Eikonal	74.08 [73.72, 74.57]	4.305 [4.300, 4.312]	20.21 [19.99, 20.44]	51.41 [51.06, 51.88]	1.842 [1.491, 2.273]
SIREN-128-LowOmega	43.54 [42.24, 45.02]	3.771 [3.744, 3.804]	12.46 [12.17, 12.84]	28.46 [27.63, 29.39]	1.795 [1.457, 2.195]
SIREN-128-Jacobian	74.29 [73.83, 74.91]	4.308 [4.302, 4.315]	20.25 [19.96, 20.53]	51.58 [51.17, 52.22]	1.864 [1.475, 2.293]

The composite RFI (Table 6) cleanly separates the models. The three $\omega_0 = 30$ models achieve negative RFI values, while LowOmega yields a positive RFI of +0.920 with non-overlapping confidence intervals.

Table 6: Composite fidelity indices for GLAS (mean [95% CI]).

Model	GeoIndex	LatentIndex	RFI
SIREN-128	-0.131 [-0.375, 0.142]	-0.475 [-0.611, -0.368]	-0.303 [-0.408, -0.192]
SIREN-128-Eikonal	-0.147 [-0.415, 0.159]	-0.444 [-0.552, -0.352]	-0.296 [-0.404, -0.192]
SIREN-128-LowOmega	0.462 [0.066, 0.826]	1.379 [1.199, 1.522]	0.920 [0.779, 1.070]
SIREN-128-Jacobian	-0.184 [-0.441, 0.084]	-0.460 [-0.595, -0.359]	-0.322 [-0.417, -0.221]

5.2.3 Robustness and Sensitivity

Deterministic differences: For every pair of GLAS models, the differences in RFI were identical across all 20 images, rendering Wilcoxon p -values inapplicable. This deterministic behaviour underscores that representational superiority is a structural property of the architecture, not a statistical artefact.

RFI weight sensitivity: Varying the GeoIndex/LatentIndex weighting from 0.3/0.7 to 0.7/0.3 did not alter the separation between LowOmega and the other models (all $\Delta\text{RFI} < -0.97$).

Spearman correlations: The composite RFI exhibited the strongest correlation with Dice ($r \approx -0.78$ to -0.86 , $p < 0.001$ after Holm correction). However, LowOmega’s RFI lies far outside the range of the other models despite only a 0.009 difference in Dice, confirming that RFI captures variance beyond Dice.

5.3 Summary of Findings Across Datasets

Both experiments support the central hypothesis: models with statistically equivalent reconstruction performance can exhibit markedly different representational fidelity. In DRIVE, SIREN-128 and SIREN-112 produced indistinguishable Dice, HD95, and ASD, yet differed in latent richness and composite RFI with large effect sizes. In GLAS, the LowOmega model achieved equivalent Dice to the $\omega_0 = 30$ variants but showed a 41% reduction in effective rank and a large positive shift in RFI. These differences are consistent across anatomical domains, model perturbations, and statistical analyses.

6. Discussion

6.1 Interpretation of Findings

The results reveal a clear dissociation between reconstruction and representational fidelity. In the GLAS experiment, the $\omega_0 = 15$ SIREN variant reconstructs gland boundaries with a clinically acceptable Dice of 0.986, yet internally encodes a substantially impoverished representation [1]. Effective rank drops by over 40%, and the composite RFI shifts from negative to positive. This suggests that the network learns a compressed representation that is sufficient for the binary reconstruction task but discards finer anatomical details.

Lowering ω_0 restricts the frequency bandwidth of the first layer, which in turn limits the range of spatial frequencies that can be represented [24, 25]. The network compensates by projecting the necessary shape information onto a lower-dimensional manifold, resulting in fewer active feature directions and reduced spectral diversity. Importantly, this compression does not harm the Dice coefficient because the boundary location is still well approximated. However, the internal representation loses richness that could be critical for tasks relying on the full SDF or the feature space, such as shape interpolation, deformation analysis, or anomaly detection [18].

In the DRIVE experiment, increasing the hidden dimension from 112 to 128 yielded only a marginal, non-significant improvement in reconstruction, yet substantially enhanced latent richness and geometric fidelity.

This suggests that capacity can influence representational quality independently of reconstruction accuracy, once a certain reconstruction threshold is met.

The deterministic nature of the observed differences in GLAS is particularly noteworthy. It implies that the effect of architectural choices on representational richness is not a statistical fluctuation but a structural property [20, 21]. This reinforces the argument that representational fidelity is a genuine, measurable attribute of continuous neural representations that cannot be inferred from reconstruction metrics alone.

6.2 Why This Matters for Medical Image Analysis

In medical imaging, segmentation accuracy is often the sole criterion for model selection. The present findings suggest that even when two models produce virtually identical segmentations, they may encode anatomical information with very different quality. For applications that go beyond binary segmentation such as shape-based morphometry, disease classification from learned features, or generative modelling of anatomy, relying exclusively on reconstruction metrics may lead to suboptimal model choices. Evaluating representational fidelity alongside reconstruction accuracy provides a more holistic assessment of model quality and can guide the development of architectures that are not only accurate but also information-rich [15].

6.3 Limitations

This study has several limitations. First, the experiments are limited to two 2D medical imaging datasets; validation on additional anatomies, modalities, and 3D data is necessary to establish broader generalisability. Second, only SIREN-based architectures were examined; other continuous representation families (e.g., ReLU MLPs with positional encoding [24], occupancy networks [3]) may exhibit different representational behaviours. Third, the clinical relevance of reduced representational fidelity was not directly evaluated through downstream task performance; future work should link the proposed fidelity metrics to specific clinical applications. Fourth, the composite RFI relies on a particular weighting of geometric and latent components; while sensitivity analyses indicate robustness, the optimal combination may be task-dependent. Despite these limitations, the core finding that statistically equivalent Dice can mask substantial representational differences provides a compelling proof-of-concept and motivates broader investigation.

6.4 Future Directions

Future work should extend the framework to 3D anatomical structures, evaluate its sensitivity to different training strategies, and investigate the relationship between representational fidelity and performance on downstream tasks. Incorporating techniques such as variational information bottleneck [31] or disentangled representation learning may help design CNRs that explicitly optimise both reconstruction and representational fidelity.

7. Conclusion

This paper has presented the first controlled investigation demonstrating that continuous neural representations with statistically identical reconstruction scores can possess fundamentally different internal representations. Through experiments on retinal vessels and glandular structures, it has been shown that models matched in Dice, HD95, and ASD can differ substantially in geometric fidelity, latent representation richness, representational similarity, and a composite Representational Fidelity Index. These differences are deterministic with respect to random initialisation and robust to variations in the composite index formulation. The results support the hypothesis that reconstruction fidelity and representational fidelity are distinct, complementary concepts. Evaluating continuous neural representations solely through reconstruction accuracy may overlook

meaningful differences in the encoded anatomical information, and incorporating representational fidelity provides a more complete picture of model quality. These findings encourage a paradigm shift toward representation-centred evaluation in medical image analysis, ultimately contributing to the development of more reliable and information-rich neural representations for clinical applications.

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